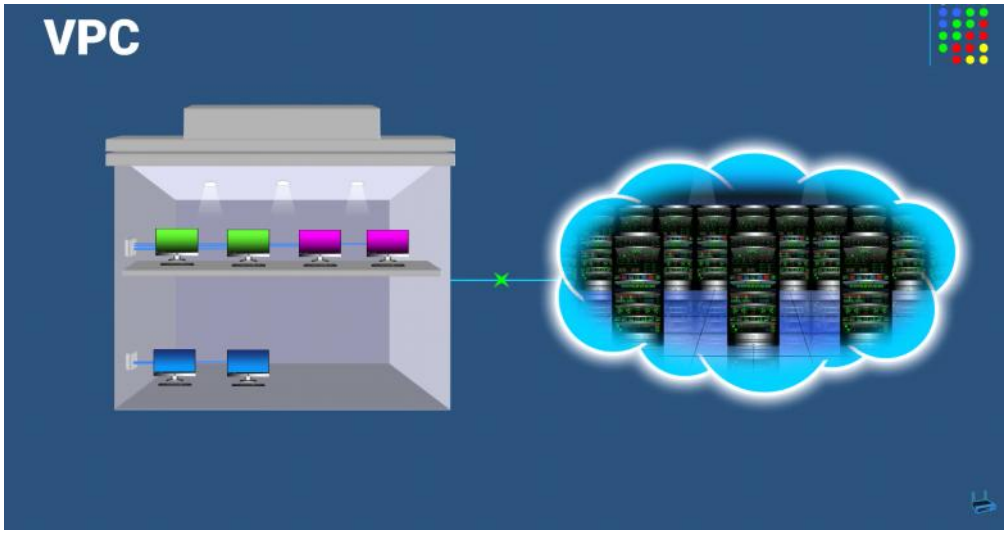
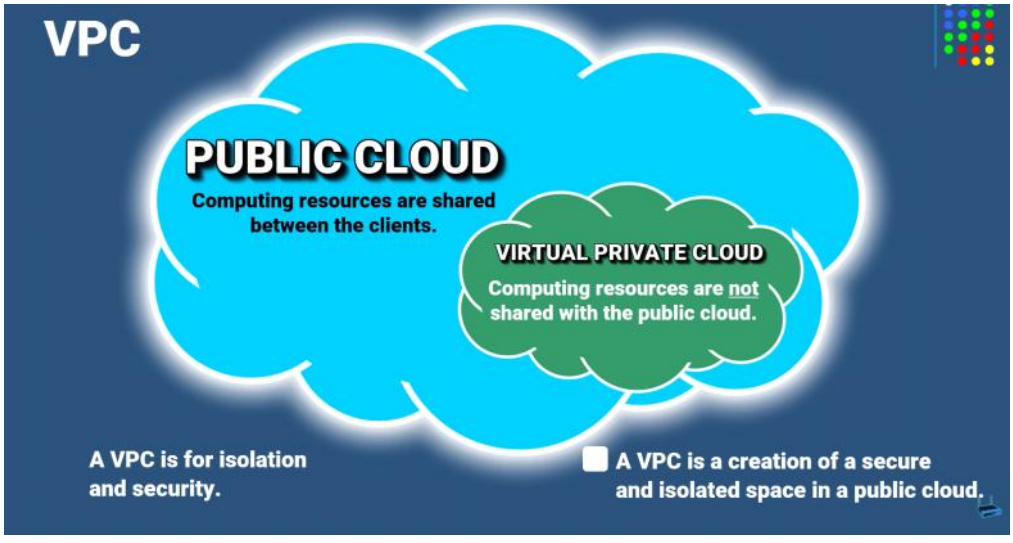
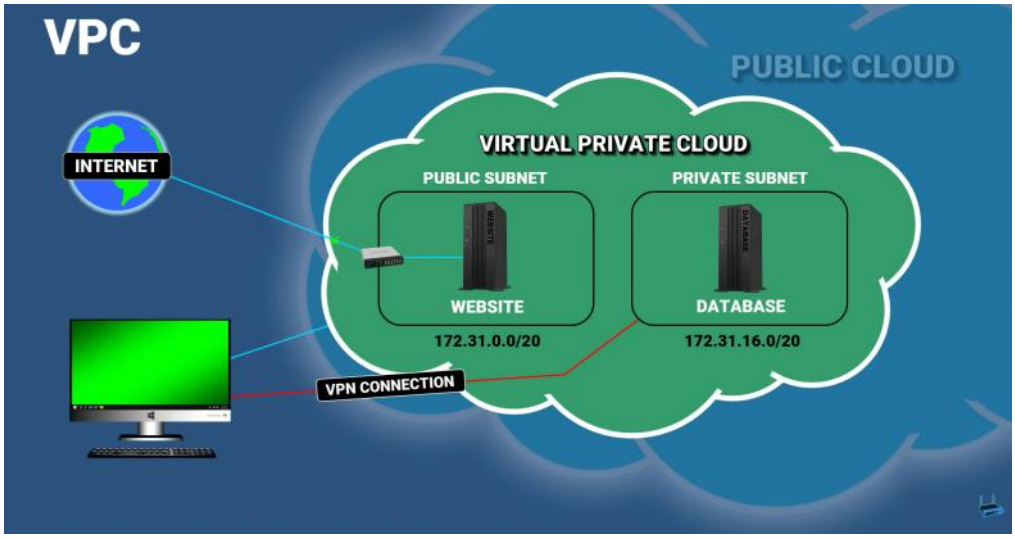
Virtual Private Cloud (VPC)

Monday, April 6, 2026 8:42 PM

A Virtual Private Cloud (VPC) is an isolated, private network environment created within a public cloud provider's infrastructure. While a public cloud shares resources across many different clients, a VPC offers a dedicated, secure space, functioning like a private building for a single tenant within a larger apartment complex.







Key Characteristics and Benefits:

Security and Isolation: By creating a logically isolated network, sensitive data and applications are separated from other public cloud users.

Flexibility and Control: Users can configure their own network settings, including subnets, route tables, and security groups.

Granular Traffic Management: You can create both public subnets (accessible via the internet using gateways) and private subnets (hidden from the public, such as for databases) to maintain different levels of security.

Scalability: VPCs allow businesses to dynamically expand or modify their network infrastructure as their needs grow.

Cost-Effectiveness: Using a VPC eliminates the heavy capital expenditure and maintenance costs associated with building and running physical, on-premises server hardware.